

Unsupervised Camouflaged Object Detection with Dual-Eigenvector Spectral Pseudo-Labeling and Contrastive Refinement

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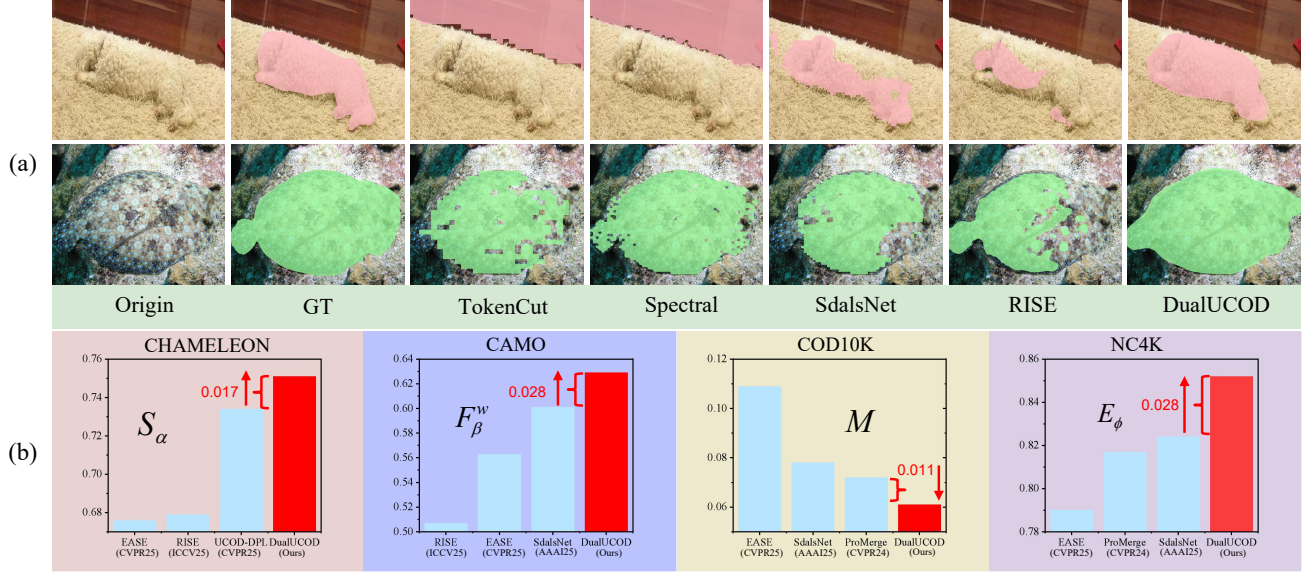


Figure 1. Qualitative and quantitative comparisons on existing unsupervised COD methods. (a) Qualitative results: our method generates more accurate masks for camouflaged objects, camouflaged objects masks are highlighted in pink and green. (b) Quantitative evaluation across different datasets: our approach outperforms SdalsNet by 2.8% in terms of the F_β^w on the CAMO dataset, demonstrating its superior segmentation capability.

Abstract

Unsupervised Camouflaged Object Detection (UCOD) aims to identify objects concealed in their surroundings without relying on pixel-level labels. Existing methods rely solely on post-processing of DINO high-dimensional features to generate pseudo labels for training. However, these methods suffer from two major limitations: 1) these pseudo labels they easily generate contain excessive noise, causing the model to learn substantial incorrect information. 2) Although pseudo-label supervision allows the model to un-

derstand the task, it remains insufficient for generating fine-grained segmentation of the camouflaged objects. To address these issues, we propose DualUCOD, a novel UCOD method based on dual-branch contrastive learning that effectively detects camouflaged objects without pixel-level labels. Specifically, we propose the Dual-Eigenvector Spectral Pseudo-Labeling (DESPL) strategy, which fuses semantic and local color consistency into an affinity matrix. We then compute the eigenvectors of its normalized graph Laplacian and generate high-quality pseudo labels using these eigenvectors. Furthermore, we introduce a Boundary-Guided Foreground-Background Refinement (BGFBR) module that explicitly incorporates boundary information to improve segmentation accuracy. Finally, we introduce a Dual-Branch Contrastive Learning (DBCL) module that constructs positive and negative pairs from the original and augmented images, aligning

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positive representations while contrasting them against negatives to enhance camouflaged object discriminability. Extensive experiments demonstrate that DualUCOD outperforms state-of-the-art methods on different datasets in the unsupervised setting. The code is available at <https://github.com/LPZliu/DualUCOD>.

1. Introduction

Camouflaged Object Detection (COD) aims to identify objects that are visually highly blended with their surroundings and difficult to detect. Its applications span wildlife conservation (Truong et al., 2024), medical image analysis (Fan et al., 2020), and concealed defect detection (He et al., 2023a). Unlike conventional segmentation tasks (Kim et al., 2024; Xu et al., 2023; Niu et al., 2024; Huang et al., 2025), the foreground and background in COD are highly similar in appearance and lack distinctive discriminative features, significantly increasing the difficulty of model recognition.

Previous methods (He et al., 2023a; Hao et al., 2025; He et al., 2025b) have predominantly focused on the COD under the fully supervised learning. However, such models heavily rely on manually annotated pixel-level labels, and producing precise annotations for camouflaged objects is often time-consuming and labor-intensive, demanding substantial human effort. To reduce reliance on pixel-level labels, some researchers (He et al., 2023b; Chen et al., 2024b; He et al., 2023b; 2025a) have turned to weakly supervised learning to guide model training, including the use of scribble annotations, bounding box annotations, and pseudo labels generated by the Segment Anything Model (SAM) (Kirillov et al., 2023; Huang et al., 2026). Although the aforementioned methods have mitigated the dependence on pixel-level labels, they still rely on human-crafted supervisory signals to drive segmentation, which naturally constrains further progress.

Recently, some researchers (Yan et al., 2025; Zhang & Wu, 2023) have attempted to achieve unsupervised COD using the DINO by extracting features and supplementing them with simple post-processing to generate pseudo labels (see Fig. 1). However, these approaches suffer from the following two problems: (1) *The generated pseudo labels are excessively coarse and afflicted with substantial noise, thereby inducing the model to learn considerable amounts of erroneous or misleading information during training.* (2) *Although pseudo-labels provide an initial conceptual understanding for the model in the COD, they are incapable of delivering fine-grained segmentation.*

To address these problems, we propose **DualUCOD**, a novel dual-branch contrastive learning framework for Unsupervised Camouflaged Object Detection (UCOD). Firstly, we

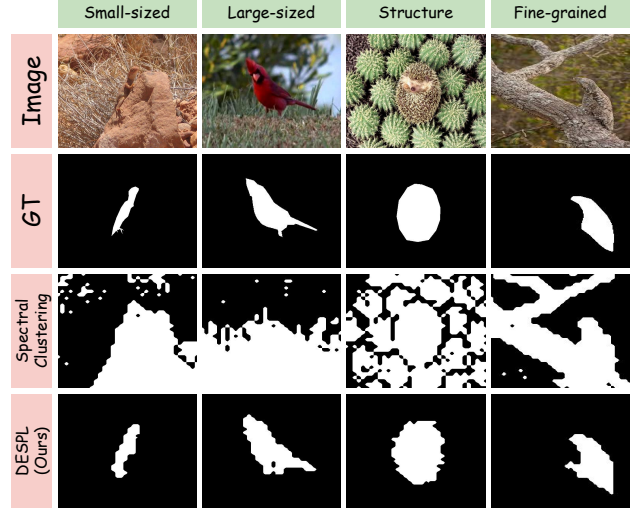


Figure 2. Visualizing the pseudo labels generated by our method in comparison with those produced by spectral clustering. Both approaches rely on features extracted from DINOv2 (Oquab et al., 2023), and the comparison is conducted across diverse scenarios, including *small-sized objects*, *large-sized objects*, *regions with structurally similar appearances*, and *complex textures prone to over-segmentation (excessive fine-grained splitting)*.

propose a Dual-Eigenvector Spectral Pseudo-Labeling (DE-SPL) strategy. By constructing an affinity matrix that integrates semantic and color cues, with semantic cues providing object-level discrimination and color cues acting as a low-level regularizer to refine boundaries and mitigate edge leakage caused by over-smoothed DINO features, we compute the eigenvectors of the normalized graph Laplacian and leverage eigenvectors to generate high-quality pseudo labels. We further propose multi-augmentation consistency fusion and adaptive dual-eigenvector fusion to improve pseudo-label quality. As illustrated in Fig. 2, our method produces higher-quality pseudo labels with less noise. To bridge the gap between pseudo labels and fine-grained segmentation, we propose a novel Boundary-Guided Foreground-Background Refinement (BGFBR) module. Unlike previous approaches that rely heavily on semantic information from the RGB domain while overlooking inherent boundary information, our method leverages it to explicitly refine the foreground and background segmentation. Finally, to improve camouflaged object discriminability, we propose a Dual-Branch Contrastive Learning (DBCL) module. By constructing positive-negative samples from the different features, we pull positive samples closer in the feature space while pushing negative samples apart, which forces the model to amplify subtle cues that distinguish camouflaged objects from similar backgrounds, thereby guiding the model to focus on discriminative characteristics of camouflaged objects. Extensive experiments demonstrate that DualUCOD outperforms existing UCOD methods.

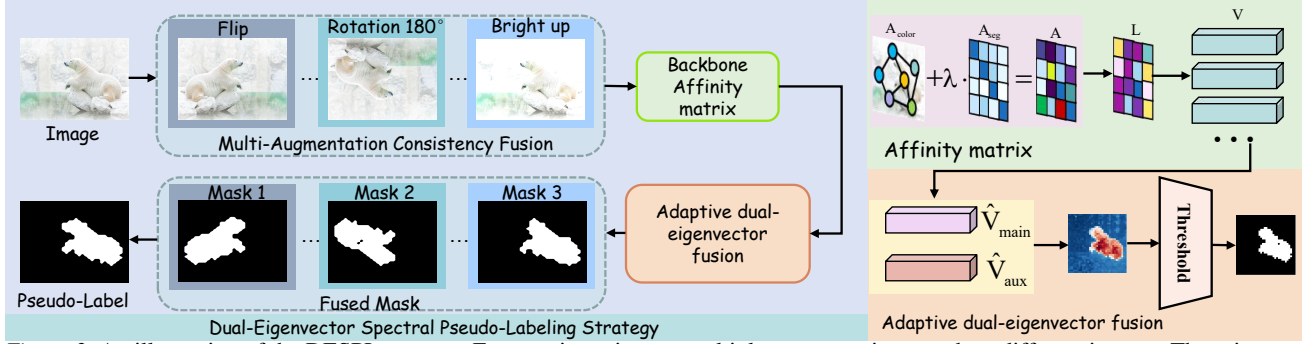


Figure 3. An illustration of the DESPL strategy. From an input image, multiple augmentations produce different images. These images are utilized to construct affinity matrices, encompassing both semantic and color affinities, leading to the formation of the Laplacian matrix. The decomposition of this matrix generates eigenvectors, which are then processed to produce the final pseudo labels.

Our contributions are summarized as follows:

- (1) We present the DualUCOD for the UCOD task, featuring a novel DESPL strategy that constructs a normalized graph Laplacian from the affinity matrix and leverages its two leading eigenvectors to generate high-quality pseudo labels for model training.
- (2) To mitigate the discrepancy between pseudo labels and fine-grained segmentation, we propose the BGFBR module, which leverages boundary information of camouflaged objects to explicitly guide the refined separation of foreground and background, thereby significantly improving segmentation accuracy.
- (3) We design the DBCL module that constructs multi-view positive-negative samples, pulling positive samples closer and pushing negative samples apart in feature space to better understand camouflaged objects.
- (4) Extensive experiments demonstrate the effectiveness of our proposed method across different COD datasets.

2. Related Work

2.1. Camouflaged Object Detection

Camouflaged object detection has achieved remarkable breakthroughs owing to the rapid advancement of deep learning (He et al., 2023a; Xiao et al., 2023; 2024; Xu et al., 2024; Liu et al., 2025; Hao et al., 2025; He et al., 2025b; Yu et al., 2026; Ye et al., 2026). For example, FEDER (He et al., 2023a) leveraged feature decomposition and edge reconstruction to achieve COD. RUN (He et al., 2025b) proposed a theoretical framework that incorporates reversible strategies in both the mask and RGB domains to achieve high-precision segmentation. Some researchers have explored training COD models under weak supervision, utilizing label forms such as scribbles (He et al., 2023b), bounding boxes (Zhang et al., 2024), points (Chen et al., 2024a), and pseudo labels (He et al., 2025a). CRNet (He et al., 2023b)

proposed a novel scribble-based learning network that leverages low-level contrasts to expand scribbles into broader camouflaged regions and uses semantic cues to refine object boundaries. SEE (He et al., 2025a) proposed using the SAM to generate pseudo labels by leveraging coarse masks from the teacher model as prompts for training COD tasks. Although the aforementioned methods have contributed modestly to the advancement of the COD tasks, they remain dependent on human-annotated supervisory signals, incurring substantial annotation costs. In contrast, we propose a novel UCOD method based on contrastive learning that generates high-quality pseudo labels to guide model training without any human annotations, enabling fine-grained segmentation.

2.2. Unsupervised Object Segmentation

Unsupervised semantic segmentation (Kim et al., 2024; Niu et al., 2024) is a pixel-level task that requires no manual annotation, aiming to group image pixels into semantically coherent regions automatically. EAGLE (Kim et al., 2024) focused on object-centric representation learning for unsupervised semantic segmentation via a spectral method leveraging deep feature similarities and color affinity. Compared with EAGLE, we propose an adaptive fusion of high-level semantics and low-level color cues, optimize eigenvector selection for camouflaged scenes, and employ a series of strategies to improve pseudo-label quality. U2Seg (Niu et al., 2024) proposed a universal unsupervised image segmentation method that generates dynamic pseudo labels through self-supervised model clustering. Although U2Seg performs well in general unsupervised image segmentation, it faces challenges in direct transfer to the COD due to the high similarity between foreground and background within a single image. Some UCOD methods (Yan et al., 2025; Du et al., 2025a) have also made progress. UCOD-DPL (Yan et al., 2025) proposed a dynamic pseudo-label learning framework that integrates a teacher-student model architecture. However, UCOD-DPL relies solely on similarity

Algorithm 1 Two-Stage Unsupervised Camouflaged Object Detection

input Unlabeled image set $\mathcal{D} = \{\mathbf{I}_i\}_{i=1}^N$; Number of augmentations per image T ; Total training epochs E ;

output Trained model f_θ

```

1: Stage I: Pseudo-Label Generation
2: Initialize  $\mathcal{P} \leftarrow \emptyset$ 
3: for each  $\mathbf{I} \in \mathcal{D}$  do
4:    $\mathbf{A} \leftarrow \text{BuildAffinity}(\mathbf{I}_i)$ 
5:    $\mathbf{V}_1, \mathbf{V}_2 \leftarrow \text{TopEigenvectors}(\mathbf{A}, k = 2)$ 
6:    $\mathbf{m}_i \leftarrow \text{Eig2Mask}(\mathbf{V}_1, \mathbf{V}_2)$ 
7:    $\mathbf{M} \leftarrow \text{MultiAugFusion}(\{\mathbf{m}_i\}_{i=1}^T)$ 
8:    $\mathcal{P} \leftarrow \mathcal{P} \cup \{\mathbf{M}\}$ 
9: end for
10: Stage II: Model Training
11: Initialize  $f_\theta$  with pre-trained weights
12: for epoch = 1 to  $E$  do
13:   for each batch  $\{(\mathbf{I}_j, \mathbf{M}_j)\}$  with  $\mathbf{I}_j \in \mathcal{D}, \mathbf{M}_j \in \mathcal{P}$  do
14:      $(\mathbf{y}_j^{\text{fg}}, \mathbf{y}_j^{\text{bg}}) \leftarrow f_\theta(\mathbf{I}_j)$ 
15:      $\mathcal{L} \leftarrow \mathcal{L}_{\text{total}}(\mathbf{y}_j^{\text{fg}}, \mathbf{M}_j) + \mathcal{L}_{\text{total}}(\mathbf{y}_j^{\text{bg}}, \mathbf{1} - \mathbf{M}_j)$ 
16:     Update  $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$  via backpropagation
17:   end for
18: end for
19: return  $f_\theta$ 
    
```

comparisons, producing low-quality and unreliable labels that can negatively impact model performance. EASE (Du et al., 2025a) proposed a novel paradigm that first identifies the environment before detecting camouflaged objects, thereby enhancing the ability to recognize camouflaged objects. However, EASE relies on multimodal large language models as its foundation, resulting in high computational costs and limiting its applicability in resource-constrained scenarios. Compared to the aforementioned methods, we propose a specialized method for UCOD that relies solely on the DINO model. It generates high-quality pseudo labels by integrating the affinity matrix, attains precise foreground-background segmentation via boundary information, and embeds a contrastive learning mechanism to strengthen discriminative capabilities for camouflaged objects.

3. Method

In this section, we introduce the novel framework for UCOD to tackle the problem of heavily depending on the pixel labels. We adopt a two-stage training strategy. In the first stage, the DESPL strategy is employed to generate high-quality pseudo labels (Sec. 3.1). In the second stage, the model is trained using these pseudo labels as supervisory signals. Given a camouflaged image, high-dimensional semantic features are extracted from a frozen backbone,

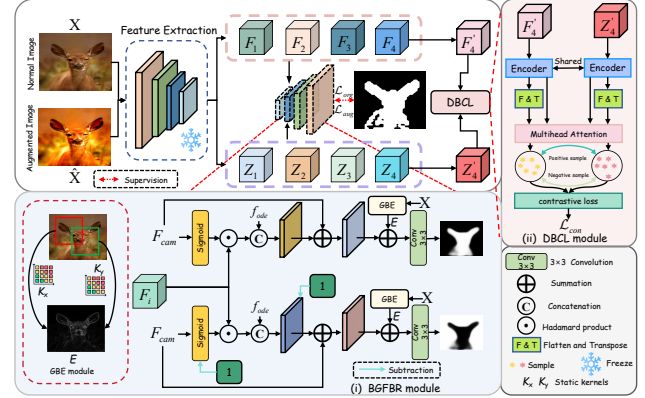


Figure 4. Framework of DualUCOD. DualUCOD comprises the frozen backbone and two dedicated modules: the BGFBR module and the DBCL module. The input firstly passes through the frozen backbone for feature extraction. In (i), we perform foreground-background segmentation for camouflaged images (the augmented branch performs the same operations as the original branch). In (ii), we enhance the model’s discriminative ability for camouflaged objects by conducting contrastive learning, thereby strengthening feature discrimination and robustness.

with only the highest-level feature adaptively refined by an Adapter module. Subsequently, the features are split into two parallel paths. One path feeds the features into the global perception module (GPM) (He et al., 2024) for multi-scale feature optimization, then passes them to the BGFBR module, producing refined foreground and background segmentation (Sec. 3.2). The other path feeds the features into the DBCL module (Sec. 3.3). Here, they undergo contrastive learning with different view features extracted from the same images after augmentation. This process enhances the model to understand the semantic consistency of camouflaged objects. We provide the pseudocode of our algorithm in Algorithm 1, which delineates the two-stage framework of our approach.

3.1. Dual-Eigenvector Spectral Pseudo-Labeling

Previous methods (Yan et al., 2025) employ overly simplistic post-processing of extracted features from DINO (Caron et al., 2021) to generate pseudo labels, resulting in low-quality masks that constrain model performance. To address this limitation, we propose the DESPL strategy (see Fig. 3), which begins by constructing an affinity matrix, from which a Laplacian matrix is derived. And eigendecomposition is then applied to obtain some eigenvectors. Furthermore, we introduce a series of refined pseudo-labeling strategies, namely multi-augmentation consistency fusion and adaptive dual-eigenvector fusion, which are designed to collaboratively produce high-quality pseudo labels.

Affinity Matrix A. Considering the characteristics of camouflaged objects, we construct an affinity matrix that integrates semantic and color cues. Given the image

Table 1. Comparison of our method with recent methods on four datasets, with results reported separately for different backbone networks. The best two results are in **red** and **blue** fonts.

Methods	Publications	CHAMELEON				CAMO				COD10K				NC4K			
		$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$
FreeSOLO (Wang et al., 2022a)	CVPR22	0.101	0.503	0.730	0.659	0.177	0.333	0.600	0.534	0.093	0.421	0.725	0.630	0.128	0.441	0.781	0.611
DiffCut (Couairon et al., 2024)	NeurIPS24	0.245	0.374	0.592	0.537	0.173	0.431	0.701	0.573	0.160	0.379	0.628	0.591	0.134	0.531	0.727	0.671
Backbone of the DINO																	
LOST (Siméoni et al., 2021)	BMVC21	0.141	0.406	0.732	0.581	0.173	0.431	0.701	0.573	0.095	0.438	0.742	0.631	0.128	0.505	0.755	0.634
Spectral (Melas-Kyriazi et al., 2022)	CVPR22	0.178	0.471	0.678	0.622	0.210	0.487	0.673	0.603	0.169	0.391	0.614	0.588	0.143	0.559	0.733	0.679
TokenCut (Wang et al., 2022b)	CVPR22	0.134	0.508	0.742	0.652	0.160	0.506	0.714	0.641	0.100	0.474	0.739	0.661	0.101	0.608	0.802	0.721
MaskCut (Wang et al., 2023)	CVPR23	0.191	0.473	0.653	0.616	0.218	0.456	0.640	0.593	0.189	0.393	0.616	0.594	0.178	0.524	0.693	0.655
FOUND (Siméoni et al., 2023)	CVPR23	0.218	0.381	0.602	0.552	0.277	0.397	0.578	0.526	0.248	0.264	0.487	0.488	0.218	0.410	0.605	0.569
ProMerge (Li & Shin, 2024)	CVPR24	0.087	0.534	0.757	0.712	0.134	0.542	0.732	0.692	0.078	0.504	0.756	0.714	0.081	0.639	0.817	0.773
SdalsNet (Shou et al., 2025)	AAAI25	0.081	0.612	0.835	0.724	0.118	0.601	0.799	0.697	0.072	0.525	0.780	0.697	0.085	0.642	0.824	0.739
EASE (Du et al., 2025a)	CVPR25	0.105	0.550	0.765	0.676	0.166	0.563	0.737	0.653	0.109	0.514	0.732	0.673	0.108	0.633	0.790	0.728
RISE (Du et al., 2025b)	ICCV25	0.090	0.508	0.741	0.679	0.132	0.507	0.715	0.664	0.062	0.526	0.795	0.724	0.081	0.633	0.818	0.759
UCOD-DPL (Yan et al., 2025)	CVPR25	0.072	0.625	0.854	0.734	0.108	0.621	0.801	0.706	0.059	0.577	0.822	0.727	0.074	0.680	0.851	0.761
DualUCOD (Ours)	-	0.064	0.643	0.864	0.751	0.099	0.629	0.820	0.723	0.061	0.535	0.795	0.704	0.068	0.671	0.852	0.764
Backbone of the DINOv2																	
TokenCut (Wang et al., 2022b)	CVPR22	0.088	0.494	0.705	0.682	0.144	0.439	0.655	0.620	0.088	0.373	0.649	0.619	0.108	0.492	0.699	0.664
MaskCut (Wang et al., 2023)	CVPR23	0.115	0.531	0.719	0.687	0.194	0.494	0.658	0.630	0.190	0.352	0.572	0.574	0.177	0.503	0.668	0.646
FOUND (Siméoni et al., 2023)	CVPR23	0.104	0.490	0.745	0.659	0.172	0.373	0.622	0.558	0.100	0.386	0.675	0.605	0.131	0.455	0.693	0.621
VoteCut (Wang et al., 2023)	CVPR24	0.145	0.546	0.735	0.674	0.170	0.522	0.702	0.637	0.092	0.534	0.782	0.690	0.097	0.649	0.818	0.739
ProMerge (Li & Shin, 2024)	CVPR24	0.085	0.567	0.787	0.741	0.142	0.502	0.706	0.679	0.081	0.435	0.714	0.674	0.096	0.558	0.771	0.726
EASE (Du et al., 2025a)	CVPR25	0.044	0.741	0.899	0.819	0.098	0.684	0.831	0.749	0.040	0.656	0.866	0.773	0.056	0.735	0.884	0.800
RISE (Du et al., 2025b)	ICCV25	0.050	0.720	0.884	0.822	0.109	0.610	0.787	0.734	0.049	0.600	0.840	0.763	0.061	0.705	0.868	0.805
DualUCOD (Ours)	-	0.041	0.754	0.918	0.827	0.083	0.702	0.852	0.771	0.042	0.635	0.856	0.761	0.052	0.745	0.896	0.808

$\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ from the COD training set, we use the DINOv2 (Oquab et al., 2023) to extract the feature map $\mathbf{H} \in \mathbb{R}^{h \times w \times D}$, where D denotes the feature dimension. After obtaining the flattened feature map $\mathbf{H}' \in \mathbb{R}^{N \times D}$, we formulate the affinity matrix by constructing its semantic component $\mathbf{A}_{\text{seg}} \in \mathbb{R}^{N \times N}$ as

$$\mathbf{A}_{\text{seg}, i, j} = \frac{\mathbf{H}'_i \cdot \mathbf{H}'_j}{\|\mathbf{H}'_i\| \|\mathbf{H}'_j\|}, \quad (1)$$

where cosine similarity is used to compute this semantic matrix. To construct the color affinity matrix, we first downsample the input image $\mathbf{I}$ to a lower resolution $\mathbf{I}' \in \mathbb{R}^{h \times w \times 3}$. At the same time, we define the color matrix $\mathbf{A}_{\text{color}} \in \mathbb{R}^{N \times N}$ as follows:

$$\mathbf{A}_{\text{color}, i, j} = \exp \left(- \left\| \mathbf{I}'_{p_i} / 255 - \mathbf{I}'_{p_j} / 255 \right\|_2^2 / (2\sigma^2) \right), \quad (2)$$

where $p_i, p_j \in \{1, \dots, hw\}$ are linear indices of pixel positions, $\mathbf{I}'_{p_i} \in \mathbb{R}^3$ denotes the RGB value of image $\mathbf{I}'$ at the i -th pixel, and $\|\cdot\|_2$ is the Euclidean norm, $N = h \times w$ denotes the total number of patch in an image of height h and width w , and σ is a hyperparameter controlling the color similarity. Moreover, due to the high similarity between camouflaged object and the background, which often results in ambiguous feature representations, we introduce a tunable hyperparameter λ (we set it to 0.1) into the color affinity matrix and fuse it with semantic information to obtain an affinity matrix that enforces local spatial consistency and mitigates semantic feature leakage across object boundaries.

$$\mathbf{A} = \mathbf{A}_{\text{seg}} + \lambda \mathbf{A}_{\text{color}}. \quad (3)$$

Laplacian Matrix $\mathbf{L}$. Formally, the Laplacian matrix is expressed as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D}$ is a diagonal matrix with $\mathbf{D}(i, i) = \sum_{j=1}^N \mathbf{A}(i, j)$. The symmetric normalized Laplacian matrix $\mathbf{L}_{\text{sym}}$ is defined as $\mathbf{L}_{\text{sym}} = \mathbf{D}^{-\frac{1}{2}} \mathbf{L} \mathbf{D}^{-\frac{1}{2}}$. Then, we extract two eigenvectors via eigendecomposition of $\mathbf{L}_{\text{sym}}$.

Adaptive Dual-Eigenvector Fusion. $\mathbf{V}_1, \mathbf{V}_2 \in \mathbb{R}^N$ denote two eigenvectors of the symmetric normalized Laplacian matrix $\mathbf{L}_{\text{sym}}$. To select a more discriminative embedding for foreground-background separation, we compute the Shannon entropy (Bromiley et al., 2004) of each eigenvector, as low-entropy eigenvectors yield more focused and boundary-aware activations, leading to more reliable foreground-background separation. First, we normalize the eigenvectors $\mathbf{V}_k$:

$$\hat{\mathbf{V}}_k = \frac{\mathbf{V}_k - \min(\mathbf{V}_k)}{\max(\mathbf{V}_k) - \min(\mathbf{V}_k)}, k = 1, 2. \quad (4)$$

Then, we estimate its histogram-based entropy:

$$\mathcal{H}(\hat{\mathbf{V}}_k) = - \sum_{b=1}^B p_{k,b} \log(p_{k,b}), \quad (5)$$

where $p_{k,b}$ is the probability of $\hat{\mathbf{V}}_k$ falling into the b -th bin. We select the eigenvector with lower entropy as the cue:

$$(\hat{\mathbf{V}}_{\text{main}}, \hat{\mathbf{V}}_{\text{aux}}) = \begin{cases} (\hat{\mathbf{V}}_2, \hat{\mathbf{V}}_1), & \text{if } \mathcal{H}(\hat{\mathbf{V}}_2) < \mathcal{H}(\hat{\mathbf{V}}_1), \\ (\hat{\mathbf{V}}_1, \hat{\mathbf{V}}_2), & \text{otherwise,} \end{cases} \quad (6)$$

where $\hat{\mathbf{V}}_{\text{main}}$ denotes the main eigenvector and $\hat{\mathbf{V}}_{\text{aux}}$ denotes the auxiliary eigenvector. An initial binary mask $\mathbf{m}^{\text{init}} \in$

$\{0, 1\}^N$ is obtained by thresholding $\hat{\mathbf{V}}_{\text{main}}$ using Otsu (Otsu et al., 1975) method:

$$m_i^{\text{init}} = \begin{cases} 1, & \text{if } \hat{\mathbf{V}}_{\text{main},i} > \tau, \\ 0, & \text{otherwise,} \end{cases} \quad \text{where } \tau = \text{Otsu}(\hat{\mathbf{V}}_{\text{main}}), \quad (7)$$

However, pixels near the decision boundary may be ambiguous. We define a low-confidence region $\mathcal{C}$ as

$$\mathcal{C} = \{i \mid |\hat{\mathbf{V}}_{\text{main},i} - \tau| \leq \delta\}, \quad (8)$$

$$\delta = \alpha \cdot (\max(\hat{\mathbf{V}}_{\text{main}}) - \min(\hat{\mathbf{V}}_{\text{main}})),$$

where α is set to 0.1. Within $\mathcal{C}$, we refine the prediction using the auxiliary eigenvector:

$$\mathbf{m}_i = \begin{cases} \mathbf{m}_i^{\text{init}}, & i \notin \mathcal{C}, \\ \mathbf{I}(\hat{\mathbf{V}}_{\text{aux},i} > 0.5), & i \in \mathcal{C}. \end{cases} \quad (9)$$

Finally, the output mask $\mathbf{m}$ is obtained by assembling all pixel-wise predictions $\mathbf{m}_i$.

Multi-Augmentation Consistency Fusion. To further improve the quality of pseudo labels, we apply T distinct data augmentation strategies (e.g., *rotation*, *flipping*, and *photometric adjustments*) to the input image, yielding a set of augmented masks $\{\mathbf{m}^t\}_{t=1}^T$. For each augmentation t , we apply the corresponding inverse transformation to obtain the results. The final pseudo-label $\mathbf{M}$ is then obtained by averaging these spatially aligned results:

$$\mathbf{M} = \frac{1}{T} \sum_{t=1}^T \mathcal{T}_t^{-1}(\mathbf{m}^t), \quad (10)$$

where $\mathcal{T}_t^{-1}$ denotes the inverse mapping corresponding to the t -th augmentation transformation.

3.2. Boundary-Guided Foreground-Background Refinement

Constrained by the unsupervised setting, pixel-level labels and edge maps cannot be used for supervision, making it difficult to recover detailed structures. Existing approaches (He et al., 2023a; 2025b; Yan et al., 2025) commonly overlook the intrinsic boundary information from RGB images. Motivated by this, we propose the BGFBR module that leverages boundary information extracted from RGB images to guide the model toward producing finer segmentation (see Fig. 4(i)). Given the images $\mathbf{X} \in \mathbb{R}^{H \times W \times 3 \times B}$, the backbone extracts four feature maps $F_i \in \mathbb{R}^{h \times w \times C \times B}$, $i = 1, 2, 3, 4$. The feature F_4 is first refined by a lightweight adapter module $\Psi(\cdot)$ to obtain the enhanced representation F'_4 , which is then fed into the GPM (Gp) to effectively capture and fuse rich multi-scale contextual information, denoted as F_{cam} . This process yields the following intermediate features:

$$F'_4 = \Psi(F_4), F_{\text{cam}} = Gp(F'_4). \quad (11)$$

Moreover, we propose a gradient-guided boundary enhancement (GBE) module that uses two orthogonal static kernels (K_x, K_y) to capture horizontal and vertical gradients and fuses them into a normalized boundary map.

$$G_x = \mathbf{X} * K_x, G_y = \mathbf{X} * K_y, \quad (12)$$

where G_x denotes the horizontal gradient components and G_y denotes the vertical gradient components of the images. Subsequently, we then compute the gradient magnitude E to capture boundary information:

$$E = \sqrt{G_x^2 + G_y^2}. \quad (13)$$

To further incorporate fine-grained features, we employed the ordinary differential equation (ODE) module (He et al., 2023a) to obtain new fusion features f_{ode} .

$$f_{ode} = \text{ODE}(\text{conv1}(\text{con}(F_i, \text{Up}(F_{\text{cam}})))), \quad (14)$$

where $\text{Up}(\cdot)$ and $\text{con}(\cdot)$ denote the up-sampling operation and the concatenation operation, respectively. conv1 represents a 1×1 convolution operation.

Foreground. The coarse foreground masks cf_{gi} can be formulated as $cf_{gi} = \sigma(F_{\text{cam}}) \odot F_i$, where $\sigma(\cdot)$ denotes the sigmoid function and $\odot$ represents element-wise multiplication. Then we further improve the quality of the foreground mask rf_{gi} .

$$rf_{gi} = \text{CBR}(\text{RCAB}(\text{con}(cf_{gi}, f_{ode})) + F_{\text{cam}}), \quad (15)$$

where $\text{RCAB}(\cdot)$ denotes the residual channel attention block (Zhang et al., 2018), CBR denotes 3×3 convolution, batch normalization, and ReLU operations. To obtain a more refined foreground mask fg_i , we explicitly fuse the boundary information as $fg_i = \text{conv3}(rf_{gi} + E)$, where conv3 denotes a 3×3 convolution.

Background. Similarly, we obtain a refined background mask using the same procedure. The coarse background mask cb_{gi} is formulated as $cb_{gi} = (1 - \sigma(F_{\text{cam}})) \odot F_i$. Then we further improve the quality of the refined background mask rb_{gi} .

$$rb_{gi} = \text{CBR}(\text{RCAB}(\text{con}(cb_{gi}, f_{ode})) + (1 - \sigma(F_{\text{cam}}))). \quad (16)$$

To obtain refined background mask bg_i , we also incorporate boundary information as $bg_i = \text{conv3}(rb_{gi} + E)$.

3.3. Dual-Branch Contrastive Learning

Due to the high visual similarity between camouflaged objects and their backgrounds, the model is susceptible to distraction by non-camouflaged object regions. To address this challenge, we introduce the DBCL module, which steers the model toward authenticating the camouflaged object by

Table 2. Ablation study on key components.

Model	COD10K				NC4K			
	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$
w/o DBCL	0.048	0.619	0.837	0.754	0.054	0.733	0.885	0.804
w/o BGFBR	0.052	0.598	0.813	0.739	0.061	0.716	0.868	0.789
Ours	0.042	0.635	0.856	0.761	0.052	0.745	0.896	0.808

 Table 3. Analysis of Hyperparameters λ .

λ	COD10K				NC4K			
	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$
0.0	0.044	0.614	0.847	0.751	0.054	0.728	0.889	0.799
0.1	0.042	0.635	0.856	0.761	0.052	0.745	0.896	0.808
0.2	0.046	0.602	0.844	0.728	0.065	0.699	0.864	0.762

focusing on subtle discriminative cues and enforcing response consistency over the target region across different views (see Fig. 4(ii)). We introduce the image augmented via photometric transformations $\hat{\mathbf{X}}$. The same backbone extracts four feature maps $Z_i \in \mathbb{R}^{h \times w \times C \times B}$. The feature Z_4 , which corresponds to the highest-level output of the backbone and thus carries rich semantic information, is also refined by an adapter $\Psi'(\cdot)$ to get the new feature map Z'_4 :

$$Z'_4 = \Psi'(Z_4). \quad (17)$$

We first utilize shared encoder layers to transform the two feature maps F'_4, Z'_4 into spatial token sequences $\mathcal{F}, \hat{\mathcal{F}} \in \mathbb{R}^{L \times C}$, where L denotes the sequence length.

$$\mathcal{F} = \text{Encoder}(F'_4), \quad \hat{\mathcal{F}} = \text{Encoder}(Z'_4), \quad (18)$$

where $\text{Encoder}(\cdot)$ comprises multiple operations, *i.e.*, convolution, batch normalization, ReLU, average pooling, and flattening. Subsequently, we generate a content-aware soft spatial query $q, \hat{q}$, formulated as follows:

$$q = \sum_{i=1}^L S(\mathcal{F})_i \cdot (\mathcal{F})_i, \quad \hat{q} = \sum_{i=1}^L S(\hat{\mathcal{F}})_i \cdot (\hat{\mathcal{F}})_i, \quad (19)$$

where $S(\cdot)$ denotes the softmax operation. Then, they are separately fed into the multi-head self-attention mechanism (MHSA($\cdot$)) and the feedforward network (FFN($\cdot$)) for processing to obtain the different view features $z, \hat{z}$:

$$z = \text{FFN}(\text{MHSA}(q, \mathcal{F})), \quad (20)$$

$$\hat{z} = \text{FFN}(\text{MHSA}(\hat{q}, \hat{\mathcal{F}})). \quad (21)$$

We project and normalize the aforementioned features to obtain the final feature map $\mathbf{u}, \hat{\mathbf{u}}$.

$$\mathbf{u} = \frac{\text{Proj}(z)}{\|\text{Proj}(z)\|_2}, \quad \hat{\mathbf{u}} = \frac{\text{Proj}(\hat{z})}{\|\text{Proj}(\hat{z})\|_2}, \quad (22)$$

where $\text{Proj}(\cdot)$ denotes Linear-ReLU-Linear operations and $\|\cdot\|_2$ denotes the L2 norm. Finally, we construct positive and negative samples from the above features for contrastive learning, where a positive pair consists of the original and augmented views of the same image. The contrastive loss is defined as follows:

$$\mathcal{L}_{\text{con}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\hat{\mathbf{u}}_i^\top \mathbf{u}_i / \tau)}{\sum_{j=1}^B \exp(\hat{\mathbf{u}}_i^\top \mathbf{u}_j / \tau)}, \quad (23)$$

where τ denotes the temperature coefficient.

3.4. Loss Function

We compute the final loss function $\mathcal{L}_{\text{total}}$, which consists of two components: the segmentation term and the contrastive term. The segmentation loss $\mathcal{L}_{\text{org}}$ for the original image is defined as follows:

$$\mathcal{L}_{\text{org}} = \sum_{i=1}^4 \mathcal{L}_{\text{BCE}}(fg_i, y) + \mathcal{L}_{\text{BCE}}(bg_i, 1 - y), \quad (24)$$

where $\mathcal{L}_{\text{BCE}}$ denotes binary cross-entropy loss for supervision and y denotes the pseudo labels. Meanwhile, the loss $\mathcal{L}_{\text{aug}}$ is computed in the same manner as described above. Finally, the total loss $\mathcal{L}_{\text{total}}$ can be defined as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{org}} + \mathcal{L}_{\text{aug}} + \mathcal{L}_{\text{con}}. \quad (25)$$

4. Experiments

Implementation details. We implement our method using PyTorch on a single NVIDIA RTX H100 GPU. The backbone is frozen and initialized from DINOv2 (Oquab et al., 2023). The full model is trained for 50 epochs using the Adam optimizer with a batch size of 16 and an initial learning rate of 3×10^{-4} . All images are resized as 476×476 in both the training and testing phases.

Table 4. Comparison of inference time per image.

Method	EASE	SdalsNet	RISE	DualUCOD
Inference Time (s)	1.5	0.3	0.2	0.6

Datasets and Evaluation Metrics. We use CAMO-Train (1,000 images) (Le et al., 2019) and COD10K (3,040 images) (Fan et al., 2020) as our training datasets, and employ DINOv2 (Oquab et al., 2023) as the vision backbone to generate high-quality pseudo labels. We evaluate the performance of our model on four widely used test sets: CHAMELEON (76 images) (Skurowski et al., 2018), CAMO (250 images), COD10K (2,026 images), and NC4K (4,121 images) (Lv et al., 2021). For comprehensive assessment, we adopt four standard evaluation metrics: Structure Measure (S_α) (Fan et al., 2017), Mean E-measure (E_ϕ) (Fan et al., 2018), Weighted F-measure (F_β^w) (Margolin et al., 2014), and Mean Absolute Error (M) (Perazzi et al., 2012).

4.1. Comparative Evaluation

We select a set of unsupervised methods for comparative evaluation, including FreeSOLO (Wang et al., 2022a), Diff-

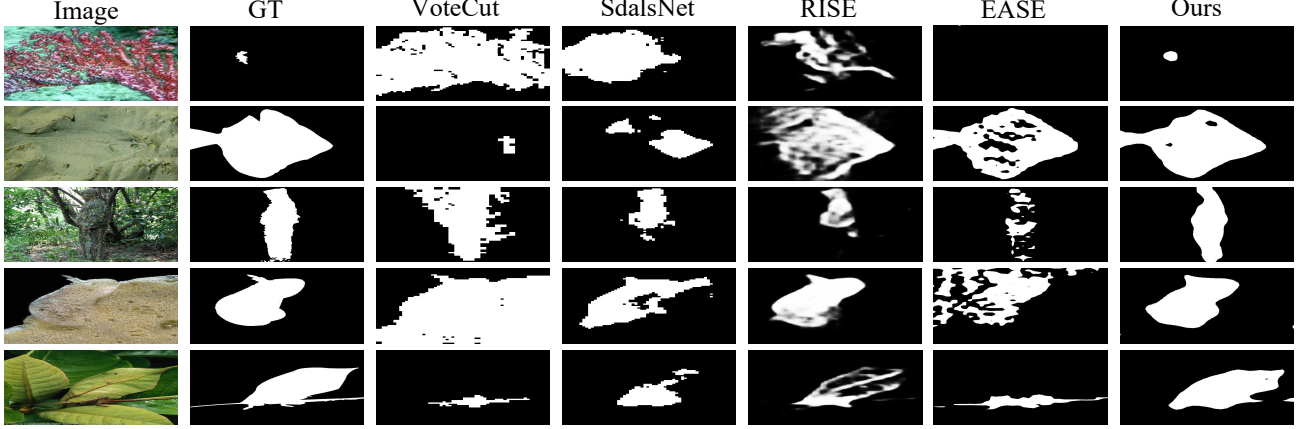


Figure 5. Visual comparison of our method with other approaches. Our method demonstrates a better understanding of camouflaged objects and produces more precise segmentation results.

Table 5. Ablation study of BGFBR submodules.

Module	COD10K				NC4K			
	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$
w/o GBE	0.050	0.603	0.823	0.743	0.055	0.728	0.881	0.799
w/o ODE	0.043	0.626	0.849	0.756	0.054	0.740	0.891	0.802
w/o RCAB	0.043	0.635	0.850	0.759	0.053	0.744	0.891	0.805
w/ BGFBR	0.042	0.635	0.856	0.761	0.052	0.745	0.896	0.808

Table 7. Impact on the number of eigenvectors k in DualUCOD.

k	COD10K				NC4K			
	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$
1	0.051	0.574	0.821	0.725	0.068	0.687	0.865	0.766
2	0.042	0.635	0.856	0.761	0.052	0.745	0.896	0.808
5	0.046	0.616	0.852	0.738	0.057	0.728	0.887	0.782

Table 9. Ablation study of the loss function.

Model	COD10K				NC4K			
	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$
w/o $\mathcal{L}_{aug}$	0.048	0.581	0.819	0.730	0.059	0.697	0.869	0.780
Ours	0.042	0.635	0.856	0.761	0.052	0.745	0.896	0.808

Cut (Couairon et al., 2024), LOST (Siméoni et al., 2021), Spectral (Melas-Kyriazi et al., 2022), TokenCut (Wang et al., 2022b), MaskCut (Wang et al., 2023), FOUND (Siméoni et al., 2023), VoteCut (Wang et al., 2023), ProMerge (Li & Shin, 2024), SdalsNet (Shou et al., 2025), EASE (Du et al., 2025a), RISE (Du et al., 2025b), and UCOD-DPL (Yan et al., 2025). All methods follow the experimental settings and data configurations reported in their original papers.

Quantitative comparison. As shown in Table 1, DualUCOD achieves competitive performance, outperforming several recent unsupervised UCOD methods on key metrics. Under the same DINO backbone setting, DualUCOD consistently improves upon SdalsNet on the CAMO with M metric decreasing by 0.019, F_{β}^w metric increasing by 2.8%, E_{ϕ} metric rising by 2.1%, and S_{α} metric improving by 2.6%. Compared with the EASE method, DualUCOD achieves

Table 6. Ablation study on DESPL strategy.

Model	COD10K				NC4K			
	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$
DualUCOD+Spectral	0.061	0.573	0.820	0.720	0.067	0.687	0.867	0.767
DualUCOD+DESPL	0.042	0.635	0.856	0.761	0.052	0.745	0.896	0.808
SINet(Fan et al., 2020)+Spectral	0.152	0.444	0.673	0.629	0.116	0.608	0.784	0.716
SINet(Fan et al., 2020)+DESPL	0.070	0.511	0.766	0.687	0.075	0.654	0.835	0.750

Table 8. Impact on the number of eigenvectors k in SINet.

k	COD10K				NC4K			
	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi} \uparrow$	$S_{\alpha} \uparrow$
1	0.108	0.413	0.667	0.618	0.112	0.552	0.753	0.682
2	0.070	0.511	0.766	0.687	0.075	0.654	0.835	0.750
5	0.111	0.486	0.719	0.663	0.091	0.647	0.816	0.746

consistent superiority across all datasets under the same DINO vision backbone. Specifically, DualUCOD outperforms EASE by 9.3%, 6.6%, 2.1%, and 3.8% in terms of the F_{β}^w metric, respectively. Compared to methods using DINOv2 as the vision backbone, our approach still achieves excellent performance on CHAMELEON, CAMO, and NC4K. Notably, on the challenging NC4K dataset, our method outperforms RISE by reducing the M metric by 0.009, improving the F_{β}^w metric by 4.0%, the E_{ϕ} metric by 2.8%, and the S_{α} metric by 0.3%. The above experiments demonstrate the effectiveness of our method in UCOD.

Qualitative comparison. As shown in Fig. 5, our method outperforms existing approaches with more accurate localization and refined segmentation of camouflaged objects, demonstrating its superior effectiveness and reliability.

Efficiency Analysis. To evaluate efficiency, we compare inference time for the image under the same settings in Table 4. DualUCOD takes only 0.6 s, which is faster than EASE, demonstrating its suitability for deployment under resource-constrained conditions.

4.2. Ablation Study

We conduct ablation experiments on the COD10K-Test and the NC4K datasets to validate our proposed components.

Effect of the DBCL. As shown in Table 2, we demonstrate the effectiveness of the DBCL module (only remove). After incorporating the DBCL module, the model shows improvements of 1.9%, 1.6%, 1.1%, and 1.2% in the E_ϕ and F_β^w metrics on the COD10K and NC4K datasets, respectively. These results suggest that DBCL module can effectively improve the model to discriminate camouflaged objects.

Effect of the BGFBR. As shown in Table 2, we validate the effectiveness of the BGFBR module. By incorporating this module, the model achieves improvements, with gains of 4.3% and 2.8% in the E_ϕ metric, and 3.7% and 2.9% in the F_β^w metric on the COD10K and NC4K. To further assess the contribution of each submodule in the BGFBR module, we conduct an ablation study. As shown in Table 5, GBE consistently achieves the significant improvements, boosting performance by 3.3% and 1.5% in E_ϕ metric, and by 3.2% and 1.7% in F_β^w metric on the COD10K and NC4K, respectively. These results demonstrate that boundary information extracted from RGB images effectively guides the model to produce more accurate segmentation.

Effect of the Hyperparameters λ . As shown in Table 3, we test $\lambda = 0.0, 0.1$, and 0.2 . The model performs best at $\lambda = 0.1$, indicating that moderate color-based constraints help align object boundaries, while excessive reliance on color degrades segmentation due to the high foreground-background similarity. This further demonstrates that effectively integrating high-level semantic cues with low-level color cues is crucial for segmenting camouflaged objects.

Effect of the DESPL and the Hyperparameters k . As shown in Table 6, we conduct comparative experiments using pseudo labels generated by Spectral and DESPL, respectively, on both our DualUCOD framework and the SINet (Fan et al., 2020) model. The results demonstrate that DESPL produces higher-quality pseudo labels, consistently yielding excellent performance on both COD10K and NC4K datasets across all models. Notably, when equipped with DESPL pseudo labels, SINet achieves remarkable improvements of 5.8% and 3.4% in the S_α metric on COD10K and NC4K, respectively. Meanwhile, we perform an ablation study on the number of eigenvectors k , setting k to 1, 2, and 5, and evaluate both DualUCOD and SINet. As shown in Tables 7 and 8, the best performance is achieved when $k = 2$. Notably, for SINet, using $k = 2$ yields S_α metric improvements of 6.9% and 2.4% over $k = 1$ and $k = 5$, respectively, on COD10K and NC4K. Notably, DualUCOD exhibits strong robustness to variations in the k , owing to its superior performance and well-designed architecture.

Effect of the loss function $\mathcal{L}_{aug}$. To further verify the

effectiveness of the loss term $\mathcal{L}_{aug}$, we conduct an ablation study in Table 9. The results show that removing $\mathcal{L}_{aug}$ leads to a noticeable performance drop, demonstrating that co-training segmentation on different views enables the model to better capture cues critical for camouflaged objects.

5. Conclusions

This paper proposes a novel UCOD method to detect camouflage objects. Firstly, we design a pseudo-label generation strategy named DESPL which improves pseudo-label quality. Secondly, we introduce the BGFBR module, which effectively integrates boundary information to refine foreground and background segmentation. Additionally, we introduce the DBCL module, which constructs positive-negative samples across different views for contrastive learning. Extensive experiments demonstrate that our proposed method achieves outstanding performance.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. You *can* have an appendix here.

A.1. A Dual-Eigenvector Pseudo-Labeling Strategy for Foreground-Background Segmentation Based on Spectral Clustering Theory

We provide a rigorous theoretical justification showing that, under the normalized cut (Ncut) formulation, the first and second eigenvectors of the symmetric normalized Laplacian are optimal for binary (foreground background) segmentation, whereas higher-order eigenvectors are not.

Preliminaries. Let $G = (V, E)$ be a connected undirected weighted graph with adjacency matrix $\mathbf{A}$ and degree matrix $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$. The symmetric normalized Laplacian is defined as

$$\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}. \quad (26)$$

Since G is connected, $\mathbf{L}_{\text{sym}}$ is symmetric positive semidefinite with eigenvalues

$$0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n \leq 2, \quad (27)$$

and corresponding orthonormal eigenvectors $\{\mathbf{V}_1, \dots, \mathbf{V}_n\}$.

Step 1: Characterization of $\mathbf{V}_1$. The eigenvector associated with $\lambda_1 = 0$ satisfies

$$\mathbf{L}_{\text{sym}} \mathbf{V}_1 = \mathbf{0} \iff \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2} \mathbf{V}_1 = \mathbf{V}_1. \quad (28)$$

Let $\mathbf{u} = \mathbf{D}^{-1/2} \mathbf{V}_1$. Then

$$(\mathbf{D} - \mathbf{A})\mathbf{u} = \mathbf{0}. \quad (29)$$

Since the graph is connected, the null space of the unnormalized Laplacian $\mathbf{D} - \mathbf{A}$ is one-dimensional and spanned by the constant vector $\mathbf{1}$. Hence,

$$\mathbf{V}_1 \propto \mathbf{D}^{1/2} \mathbf{1}. \quad (30)$$

Thus, the i -th component is $|\mathbf{V}_1|_i = \sqrt{d_i}$, where $d_i = \sum_j \mathbf{A}_{ij}$ reflects the total affinity of node i to its neighbors. In camouflage segmentation, the graph is constructed such that background regions tend to be smooth and homogeneous, leading to higher degrees, whereas foreground objects are sparse or irregular, yielding lower degrees. Consequently, $\mathbf{V}_1$ exhibits systematic magnitude differences between foreground and background nodes.

Step 2: Normalized Cut as a Variational Problem. For a bipartition $(S, \bar{S})$, the normalized cut objective is defined as

$$\text{Ncut}(S, \bar{S}) = \frac{\text{cut}(S, \bar{S})}{\text{vol}(S)} + \frac{\text{cut}(S, \bar{S})}{\text{vol}(\bar{S})}. \quad (31)$$

From Shi & Malik (2000), minimizing Ncut over all discrete partitions is approximately relaxed as the following constrained optimization problem:

$$\min_{\mathbf{z} \in \mathbb{R}^n} \mathbf{z}^\top \mathbf{L}_{\text{sym}} \mathbf{z}, \quad \text{s.t. } \mathbf{z}^\top \mathbf{z} = 1, \mathbf{z}^\top \mathbf{V}_1 = 0. \quad (32)$$

This formulation explicitly excludes the trivial solution $\mathbf{V}_1$.

Step 3: Optimality of $\mathbf{V}_2$ for Binary Segmentation. By the Courant–Fischer min–max theorem,

$$\lambda_2 = \min_{\substack{\mathbf{z} \perp \mathbf{V}_1 \\ \|\mathbf{z}\|_2=1}} \mathbf{z}^\top \mathbf{L}_{\text{sym}} \mathbf{z}. \quad (33)$$

Therefore, $\mathbf{V}_2$ is the unique (up to sign) minimizer of the relaxed Ncut objective. Any feasible vector orthogonal to $\mathbf{V}_1$ that is not $\mathbf{V}_2$ yields a strictly larger objective value.

Moreover, since $\mathbf{V}_1$ has strictly positive entries, orthogonality $\mathbf{V}_1^\top \mathbf{V}_2 = 0$ implies that $\mathbf{V}_2$ must contain both positive and negative components. This sign change induces a natural bipartition:

$$S = \{i \mid \mathbf{V}_2[i] > 0\}, \quad \bar{S} = \{i \mid \mathbf{V}_2[i] < 0\}, \quad (34)$$

which corresponds to the optimal relaxed solution of the Ncut problem.

Step 4: Why Higher-Order Eigenvectors Are Suboptimal for Binary Cuts. Eigenvectors $\mathbf{V}_k$ with $k \geq 3$ satisfy

$$\mathbf{V}_k \perp \text{span}\{\mathbf{V}_1, \mathbf{V}_2\}, \quad (35)$$

and correspond to eigenvalues $\lambda_k \geq \lambda_2$. By the variational principle, any such direction yields

$$\mathbf{V}_k^\top \mathbf{L}_{\text{sym}} \mathbf{V}_k = \lambda_k \geq \lambda_2, \quad (36)$$

implying a strictly worse normalized cut value than that achieved by $\mathbf{V}_2$.

From a structural perspective, $\mathbf{V}_k$ for $k \geq 3$ necessarily exhibits additional sign changes and oscillatory behavior, reflecting finer or multi-way partitions of the graph. While such eigenvectors are essential for K -way clustering (Ng et al., 2001), they do not correspond to optimal solutions of the binary Ncut objective and therefore are not suitable for foreground-background segmentation.

In summary, $\mathbf{V}_1$ lies in the null space of $\mathbf{L}_{\text{sym}}$ and is excluded by the Ncut constraints, but its magnitude contains discriminative information relevant to foreground-background separation, while $\mathbf{V}_2$ is the unique minimizer of the relaxed binary normalized cut problem and induces the optimal bipartition via its sign structure. Higher-order eigenvectors correspond to strictly suboptimal solutions for binary cuts and encode finer-scale or multi-way structures. Consequently, the first and second eigenvectors are theoretically sufficient and optimal for foreground-background segmentation.

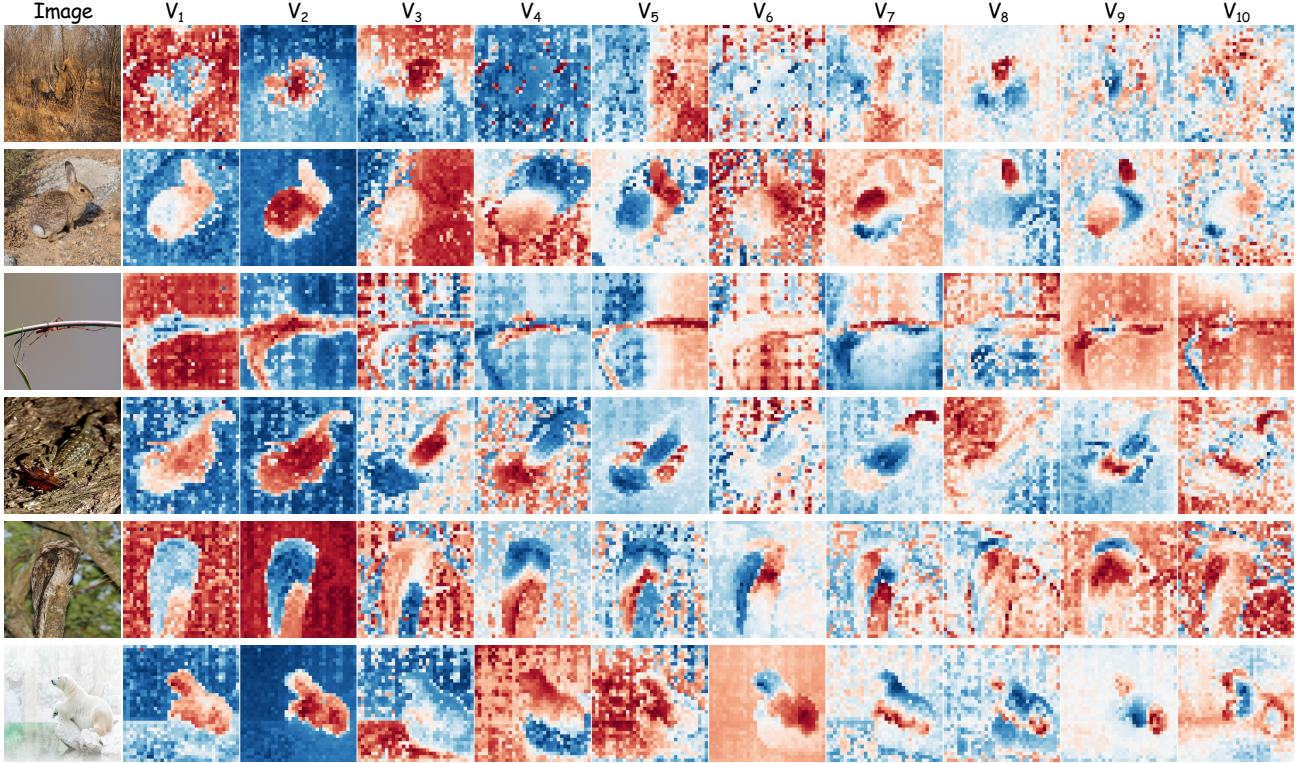


Figure 6. Visualization of the first ten eigenvectors (V_1 to V_{10}) of the normalized graph Laplacian.

Step 5: Qualitative evaluation of eigenvectors. To further verify the correctness of the aforementioned derivation, we conduct a visual analysis of the obtained eigenvectors. As shown in Fig. 6, the distribution of the first ten eigenvectors (from V_1 to V_{10}) on a 34×34 spatial grid is presented. It can be clearly observed that only the first and second eigenvectors (V_1 and V_2) exhibit coherent response regions, effectively delineating the semantic boundaries between camouflaged objects and the background. Starting from the third eigenvector, however, the response patterns gradually become fragmented and incomplete. The corresponding segmentation results commonly suffer from structural discontinuities and semantic ambiguities, making it difficult to reflect meaningful object contours. This phenomenon indicates that, in the currently constructed graph structure, the most discriminative information is primarily concentrated within the first and second eigenvectors. As shown in Fig. 6, foreground and background in camouflaged images are often ambiguous; even when V_1 exhibits an inverse intensity pattern, it still helps distinguish the two regions.

A.2. Supplement to the Method

We provide the pseudocode of our algorithm in Algorithm 1, which delineates the two-stage framework of our approach. Specifically, we first produce high-quality pseudo labels from unlabeled images and subsequently employ them as

supervision to train the model in a fully supervised setting.

A.3. Supplement to the Implementation details

We set both the h and w of the affinity matrix to 34. And we introduce five additional standard evaluation metrics to assess the quality of pseudo labels: Max F-measure (F_β^m) (Margolin et al., 2014), Adaptive E-measure (E_ϕ^a) (Fan et al., 2018), Max E-measure (E_ϕ^m) (Fan et al., 2018), Adaptive F-measure (F_β^a) (Margolin et al., 2014), and Mean F-measure F_β (Margolin et al., 2014). We employ DINOv1-ViT-B16 and DINOv2-ViT-L14 as our visual backbones. In our implementation, the color similarity bandwidth σ in Eq.(2) is set to 0.1, the number of histogram bins $\mathcal{B}$ for entropy computation in Eq.(5) is set to 50, and the number of augmentations T in Eq.(10) is set to 8.

A.4. Supplement to the Ablation Study

Effect of the Hyperparameters λ . We present an extended ablation study on the hyperparameter λ in Table 10, where we evaluate a broader set of values, specifically $\lambda=0.3, 0.5, 0.7$, and 1.0 , to analyze its impact in detail. Combined with the results in Table 3, which includes additional settings such as $\lambda=0.1$, it is evident that $\lambda=0.1$ achieves the best performance, enabling more effective recognition of camouflaged objects and generation of higher-quality pseudo labels.

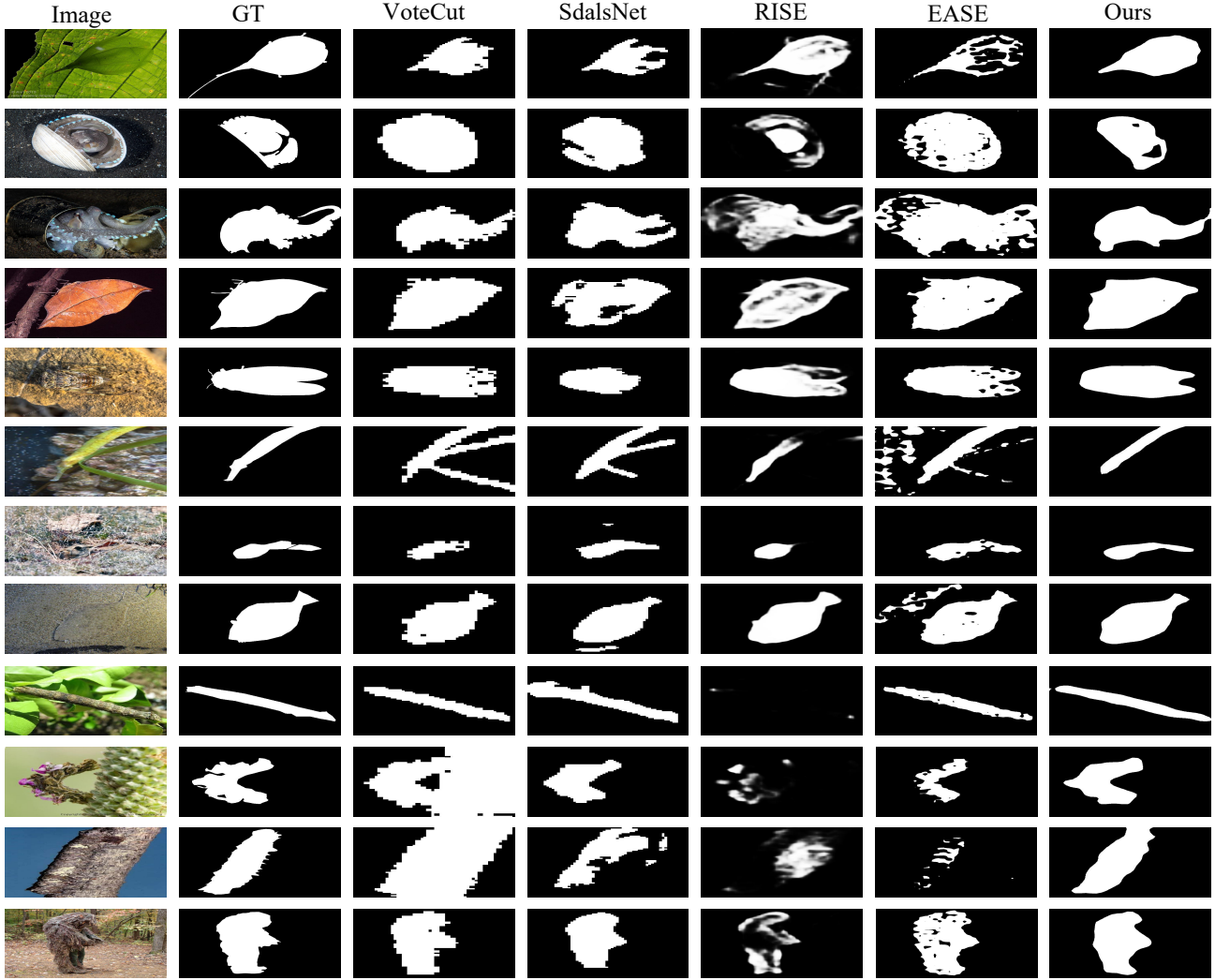


Figure 7. Extensive visual comparisons between the proposed method and various other approaches are provided.

 Table 10. Ablation study of Hyperparameters λ .

λ	COD10K				NC4K			
	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$
0.3	0.049	0.600	0.828	0.743	0.057	0.717	0.876	0.793
0.5	0.057	0.537	0.807	0.706	0.069	0.663	0.855	0.758
0.7	0.056	0.557	0.791	0.714	0.064	0.691	0.857	0.775
1.0	0.070	0.511	0.757	0.687	0.081	0.639	0.822	0.742

Effect of DESPL. We further conduct ablation studies to evaluate the refined pseudo-labeling strategies. As shown in the Table 11, this strategy effectively improves the quality of pseudo-labels and consequently improves model performance.

Effect of the More Hyperparameters. We conduct ablation studies on the key hyperparameters T , α , τ , β , and σ . As shown in the Table 12, the model achieves optimal

Table 11. Ablation study of the Adaptive Dual-Eigenvector Fusion (ADEF) and Multi-Augmentation Consistency Fusion (MACF).

Model	COD10K				NC4K			
	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$	$M \downarrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$S_\alpha \uparrow$
w/o ADEF	0.051	0.574	0.821	0.725	0.068	0.687	0.865	0.766
w/o MACF	0.055	0.593	0.812	0.736	0.056	0.719	0.873	0.794
Ours	0.042	0.635	0.856	0.761	0.052	0.745	0.896	0.808

performance when these parameters are set to 8, 0.1, 0.1, 50, and 0.1, respectively.

Quality Comparison of Pseudo Labels. We further compare the pseudo-label quality of Spectral and our method. As shown in Table 13, our pseudo-labels achieve consistently better performance than Spectral.

Qualitative Comparison. As shown in Fig. 7, we present a more comprehensive and detailed visualization of the comparison results. It is evident that compared to existing meth-

Table 12. Parameter analysis on T , α , τ , $\mathcal{B}$, and σ on the COD10K.

Metrics	T				α				τ				$\mathcal{B}$				σ			
	4	6	8	10	0.05	0.1	0.2	0.3	0.05	0.1	0.2	0.3	40	50	60	70	0.05	0.1	0.2	0.3
$M \downarrow$	0.048	0.048	0.042	0.043	0.044	0.042	0.047	0.063	0.046	0.042	0.047	0.046	0.044	0.042	0.044	0.043	0.043	0.042	0.046	0.050
$F_{\beta}^w \uparrow$	0.600	0.596	0.635	0.628	0.597	0.635	0.613	0.568	0.632	0.635	0.608	0.604	0.643	0.635	0.624	0.627	0.624	0.635	0.620	0.584
$E_{\phi} \uparrow$	0.822	0.818	0.856	0.854	0.848	0.856	0.840	0.811	0.830	0.856	0.817	0.848	0.842	0.856	0.836	0.862	0.850	0.856	0.834	0.800
$S_{\alpha} \uparrow$	0.742	0.739	0.761	0.756	0.741	0.761	0.750	0.720	0.760	0.761	0.744	0.737	0.767	0.761	0.755	0.756	0.753	0.761	0.755	0.727

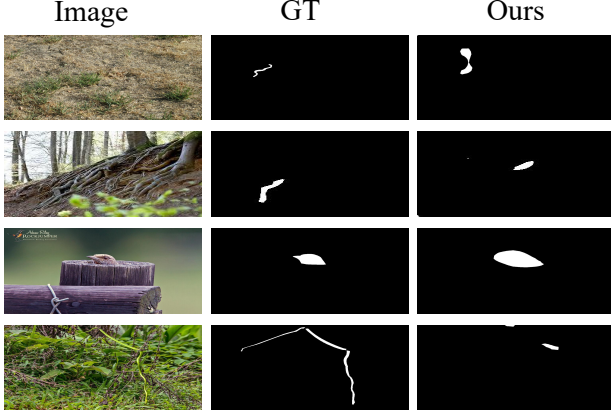


Figure 8. Visualization of Failure Cases.

high-resolution feature extraction strategies and design data augmentation techniques specifically tailored to enhance the representation and recovery of small camouflaged objects.

Table 13. Comparison of pseudo-label quality.

Method	COD10K-Train + CAMO-Train									
	$S_{\alpha} \uparrow$	$E_{\phi} \uparrow$	$F_{\beta}^m \uparrow$	$M \downarrow$	$F_{\beta}^w \uparrow$	$E_{\phi}^a \uparrow$	$E_{\phi}^m \uparrow$	$F_{\beta}^a \uparrow$	$F_{\beta} \uparrow$	
Spectral	0.658	0.708	0.626	0.168	0.571	0.710	0.710	0.626	0.624	
DESPL(Ours)	0.728	0.788	0.683	0.077	0.618	0.790	0.790	0.683	0.681	

ods, the approach proposed in this paper not only captures boundary details more accurately and achieves high-quality pixel-level segmentation when handling camouflaged objects but also gains a deeper understanding of semantic information within the scene. This enables effective identification and complete segmentation of camouflaged targets in complex backgrounds, demonstrating overall performance that significantly outperforms the comparison methods. At the same time, we also present several failure cases of this method in Fig. 8. These failures primarily occur on small objects that are both spatially tiny and visually camouflaged against complex backgrounds. The combination of limited pixel support and low appearance contrast leaves insufficient discriminative cues for reliable detection.

A.5. Limitations and Future Work

Although our method achieves notable performance gains, it remains challenged by small camouflaged objects. This limitation stems from two intertwined factors: (1) the inherently weak visual signals of such objects and (2) the low spatial resolution of DINO-derived feature maps, which further erodes fine-grained details critical for accurate localization and segmentation. In future work, we plan to explore